



Probability with Engineering Applications: Week6

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Programme

- Expectation and examples
- Change-of-variable formula
- Variance and Standard deviation
- Change-of-units formula

Expectation

Consider the following two statements:

1. The expectation $E[X]$ is the most likely value of X .
2. The expectation $E[X]$ is always positive.

Which of these statements is true?

- A. None
- B. Only 1
- C. Only 2
- D. Both

Expectation

Let X be a discrete RV with probability mass function $p(-1) = \frac{1}{5}$, $p(0) = \frac{2}{5}$ and $p(1) = \frac{2}{5}$. Let Y be equal to X^2 .

Compute $E[X]$ and $E[Y]$.

A. $E[X] = \frac{1}{5}$, $E[Y] = \frac{1}{25}$

B. $E[X] = 0$, $E[Y] = 0$

C. $E[X] = 1$, $E[Y] = 1$

D. $E[X] = \frac{1}{5}$, $E[Y] = \frac{3}{5}$

Expectation: center of mass

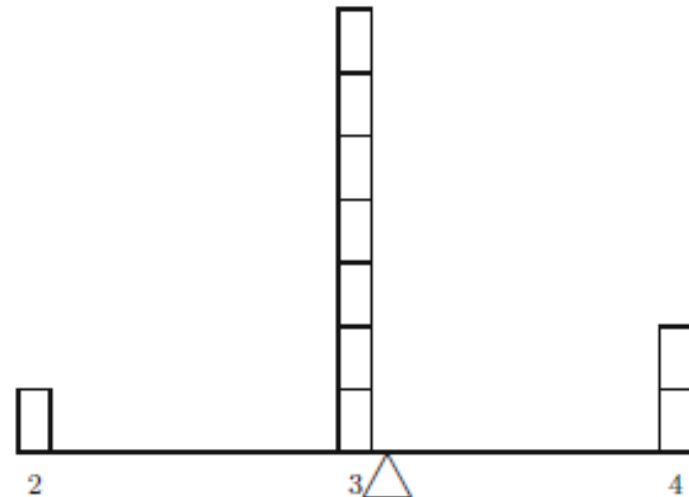
Definition:

The expectation of a discrete random variable X is defined as

$$E[X] = \sum_i a_i P(X = a_i).$$

Interpretation:

$E[X]$ is center of mass of the distribution.



Expectation: center of mass

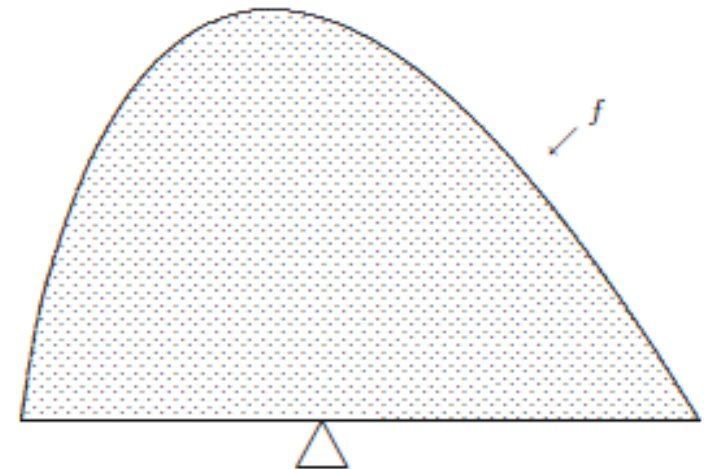
Definition:

The expectation of a continuous random variable X with pdf f is given by

$$E[X] = \int_{-\infty}^{+\infty} xf(x) dx.$$

Interpretation:

$E[X]$ is center of mass of the distribution.



Expectations of standard distributions

Distribution	Expectation	Result
$\text{Bin}(n, p)$	$\sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$	
$\text{Geo}(p)$	$\sum_{k=1}^{\infty} k p (1-p)^{k-1}$	
$\text{Pois}(\mu)$	$\sum_{k=0}^{\infty} k \frac{\mu^k}{k!} e^{-\mu}$	
$\text{Exp}(\lambda)$	$\int_0^{\infty} x \lambda e^{-\lambda x} dx$	
$N(\mu, \sigma^2)$	$\int_{-\infty}^{\infty} x \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} dx$	



Practice now

Given is a continuous random variable X whose distribution function F satisfies $F(x) = 0$ for $x < 0$, $F(x) = 1$ for $x > 1$, and $F(x) = x(2 - x)$ for $0 \leq x \leq 1$. Determine $E[X]$.



Change-of-variable formula

Change-of-variable

Given a random variable $X: \Omega \rightarrow \mathbb{R}$ and function $g: \mathbb{R} \rightarrow \mathbb{R}$, we can define a new random variable $Y: \Omega \rightarrow \mathbb{R}$ by composition.

$$Y = g(X)$$

To find the distribution function F_Y of $Y = g(X)$, we determine

$$F_Y(x) = P(Y \leq x) = P(g(X) \leq x)$$



Change-of-variable: Example

Suppose $X \sim U(-1, 2)$ and $g(x) = x^2$. Define $Y = g(X)$.

$$F_Y(x) = P(Y \leq x) = P(X^2 \leq x)$$



Change-of-variable: Example

Suppose $X \sim U(-1, 2)$ and $g(x) = x^2$. Define $Y = g(X)$.

$$F_Y(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \int_{-\sqrt{x}}^{\sqrt{x}} \frac{1}{3} d\xi = \frac{2\sqrt{x}}{3} & \text{if } 0 < x \leq 1 \\ \int_{-1}^{\sqrt{x}} \frac{1}{3} d\xi = \frac{1}{3} + \frac{\sqrt{x}}{3} & \text{if } 1 < x \leq 4 \\ 1 & \text{if } x > 4 \end{cases}$$

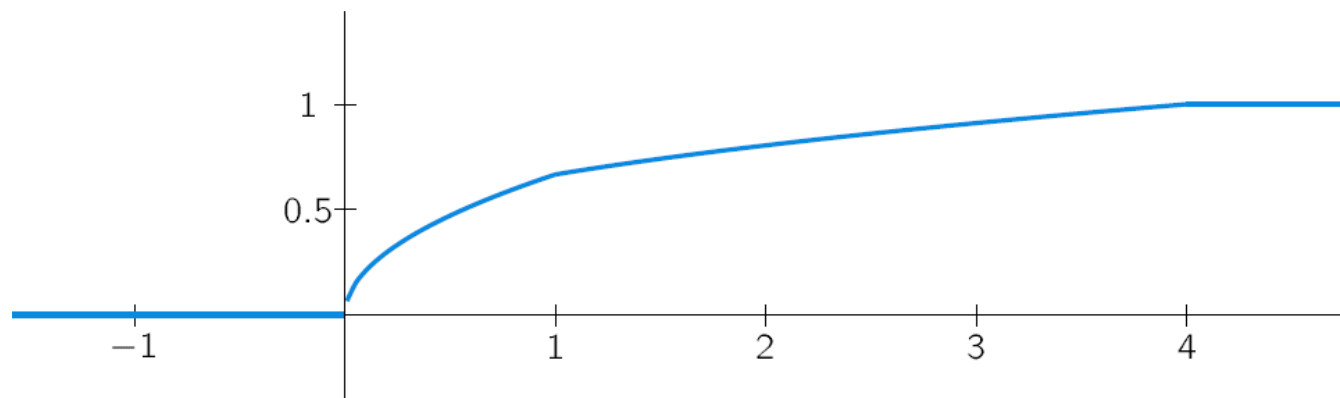
$$f_Y(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{1}{3\sqrt{x}} & \text{if } 0 < x \leq 1 \\ \frac{1}{6\sqrt{x}} & \text{if } 1 < x \leq 4 \\ 0 & \text{if } x > 4 \end{cases}$$



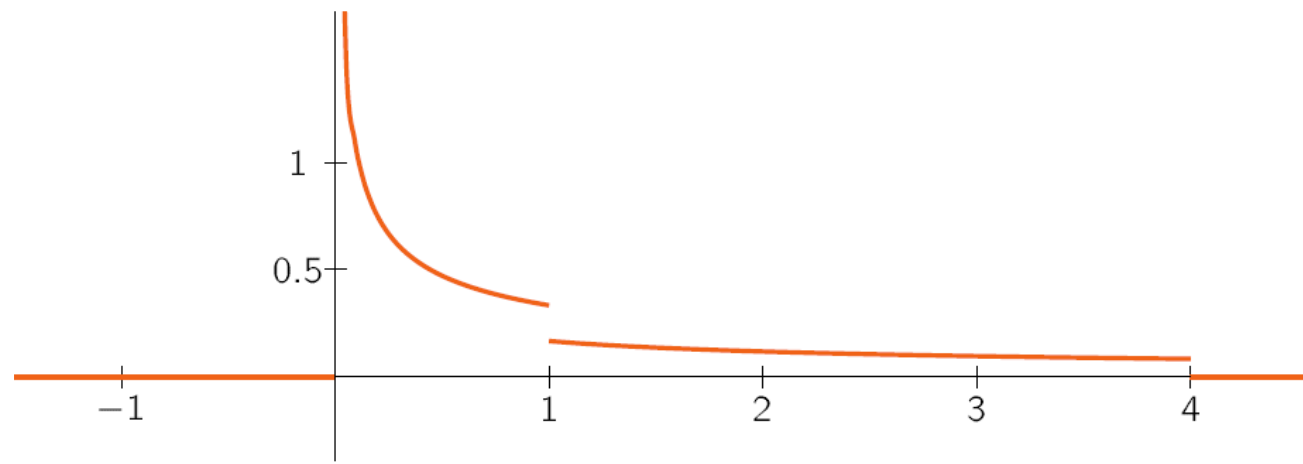
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Change-of-variable formula

Theorem:

Let X be a RV, and $g : \mathbb{R} \rightarrow \mathbb{R}$ a function.

If X is discrete, then

$$E[g(X)] = \sum_i g(a_i)P(X = a_i).$$

If X is continuous with pdf f , then

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x)f(x) dx.$$



Variance and Standard deviation

Variance

Definition:

The variance of a random variable X is defined as

$$\text{Var}(X) = E[(X - E[X])^2].$$

The standard deviation of a random variable X is defined as

$$\text{SD}(X) = \sqrt{\text{Var}(X)}.$$

For any random variable X the variance can be computed by

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$



Expectation and Variance of Standard Distributions

Distribution	Expectation	Variance
$\text{Bin}(n, p)$	np	$np(1 - p)$
$\text{Geom}(p)$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
$U(\alpha, \beta)$	$\frac{1}{2}(\alpha + \beta)$	$\frac{1}{12}(\beta - \alpha)^2$
$N(\mu, \sigma^2)$	μ	σ^2
$\text{Exp}(\lambda)$	$1/\lambda$	$1/\lambda^2$



Change-of-units formula

Change-of-units formula

Theorem:

For any random variable X and any real values r and s it holds that

$$E[rX + s] = rE[X] + s$$

$$\text{Var}(rX + s) = r^2 \text{Var}(X).$$

Remark:

This follows from the change-of-variable formula.

Do you see how?



Practice now

Let X be a random variable with $E[X] = 2$, $\text{Var}(X) = 4$. Compute the expectation and variance of $3 - 2X$.



Change-of-units formula

Consider the following two statements:

1. It is always the case that $E[X^2] \geq (E[X])^2$.
2. $\text{Var}(-2X + 1) = 4\text{Var}(X)$.

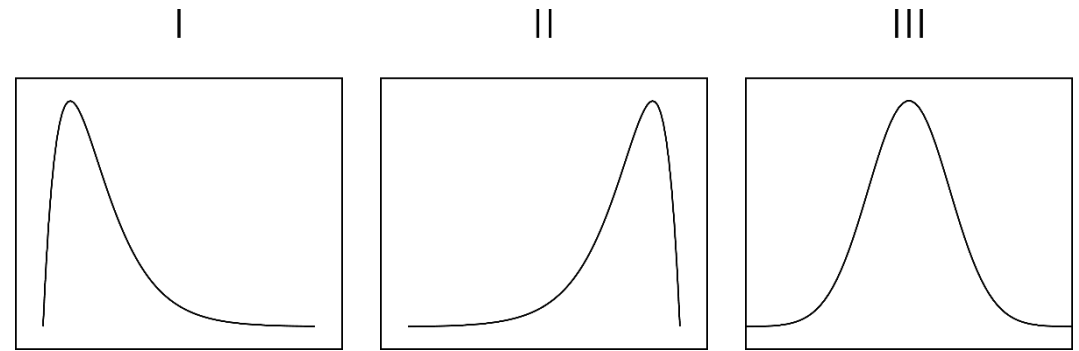
Which of these statements is true?

- A.** None
- B.** Only 1
- C.** Only 2
- D.** Both

Median vs expectation

Consider the three graphs on the right.
Let m be the median of a distribution:

$$P(X \leq m) = \frac{1}{2}$$



Which of the statements below is true?

- A.** For graph I: $m > E[X]$
- B.** For graph II: $m > E[X]$
- C.** For graph III: $m > E[X]$



See you next week